

Candidate Number

Candidate Name _____

INTERNATIONAL ENGLISH LANGUAGE TESTING SYSTEM

Academic Reading

Additional materials:

Answer sheet for Listening and Reading

Time 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this question paper until you are told to do so.

Write your name and candidate number in the spaces at the top of this page.

Read the instructions for each part of the paper carefully.

Answer all the questions.

Write your answers on the answer sheet. Use a pencil.

You must complete the answer sheet within the time limit.

At the end of the test, hand in both this question paper and your answer sheet.

INFORMATION FOR CANDIDATES

There are **40** questions on this question paper.

Each question carries one mark.

READING PASSAGE 1

You should spend about 20 minutes on Questions 1–13, which are based on Reading Passage 1

Transport and transport systems during the Roman Empire

At the height of the Roman Empire in the 2nd century AD, it was not uncommon for Romans to travel long distances by road across Europe and beyond. In fact, it was actually quite easy to travel, thanks to the empire's highly developed road and shipping networks. It has been calculated that Roman roads covered a total distance of over 400,000 kilometres, with more than 120,000 kilometres of this being of the type known as 'public roads' and a second system classified as 'military roads', both constructed by soldiers. The roads spread across the empire from Britain in the north to Morocco in the south, from Portugal in the west to Iraq in the east. Roads played a crucial role in the maintenance of the empire, allowing people and goods to move relatively swiftly. Tolls were charged for the use of many major roads, with stations where these tolls were paid often situated at bridges or at city gates.

Having the elaborate road network also meant that Roman troops could be sent rapidly to various parts of the empire in the event of emergencies, such as internal revolts or external threats. Therefore, the existence of these roads reduced the number of large and costly garrisons* needed throughout the empire.

Ordinary Romans would travel in a *raeda*, a carriage with four wooden wheels covered with iron and pulled by up to four horses. The *raeda* was the equivalent of the bus today. It had wooden benches inside for passengers, and a cloth top or sometimes no top at all. Rich Romans travelled in a *carpentum*, which was essentially a luxury vehicle for the wealthy. The *carpentum* had four wheels, a wooden arched rooftop and seats. It was pulled by many horses and was built incorporating a form of suspension to make the ride less bumpy and disagreeable. Romans also had a vehicle, called a *plaustrum*, that did not normally carry passengers—what would be considered a truck today. The *plaustrum*, constructed with a flat wooden base over four thick wheels and drawn by two oxen, was designed to carry heavy loads. It was very slow and could cover only 15 to 25 kilometres per day.

The fastest way to travel over heavily trafficked routes was by the *cursus publicus*. This was like a state-run postal service, but was also used to transport government or military officials. A certificate issued by the emperor was required in order to use the service. A series of stations with fresh horses were built at short regular intervals of approximately 12 kilometres along the major road systems. It has been estimated that a typical trip by relays of horses was made at a rate of 66 to 103 kilometres per day.

**garrison: a group of soldiers stationed in a fortress or town to defend it*

Because of their iron-shod wheels, Roman vehicles made a lot of noise and were therefore forbidden by law from entering the vicinity of big Roman cities during the day. Fortunately, many roads had stations, called mansiones, where Roman travellers could stop and break their journey, the equivalent of motorway rest areas today. These sometimes included inns where tired travellers could drink, eat and sleep. The stations were built by the government at regular intervals of 25 to 30 kilometres. One of the factors that meant such roads could be safely used for trade was the fact that they were patrolled by the Roman army. Merchants knew that they would usually be protected from bandits and highwaymen.

It was not uncommon for well-to-do Romans to travel as tourists to places relatively far away such as Greece and Egypt. For this, they would often have to sail across the Mediterranean Sea, but as there were no regular passenger services, they travelled on ships which were transporting goods. Tourists had to find a ship, get the captain's approval and negotiate a price with him. Fortunately, there were many merchant ships going back and forth regularly along established routes in the Mediterranean so tourists had little difficulty finding a ship going to their chosen destination. Most Roman tourists would travel on the open deck of the ship, sometimes with hundreds of people crowded together. Food was not included in the price of transport, so passengers had to supply their own. Certain merchant ships provided cabins at the rear that could accommodate the richer passengers. Meanwhile, the wealthiest Romans had their own ships and employed a crew to sail them, much like on big private yachts today.

Travelling by ship in Roman times was relatively fast. For example, going from Brindisium in Italy to Patrae in Greece would take about three days. Commercial navigation was suspended during particularly bad storms, when the Mediterranean was too rough and dangerous for ships to sail.

Questions 1 – 7

Do the following statements agree with the information given in Reading Passage 1? In boxes 1–7 on your answer sheet, write:

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this

1. The Roman military was responsible for building the ‘public roads’.
2. The Roman road network made it necessary to establish more garrisons.
3. Travel by raeda was usually only for short distances.
4. A carpentum was constructed with some consideration for passenger comfort.
5. People wanting to use the cursus publicus service needed official permission.
6. Mansiones were usually located where several main roads met.
7. Private businessmen constructed the mansiones.

Questions 8 – 13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes 8–13 on your answer sheet.

Roman travel by ship

- Passengers normally travelled on ships used mostly for carrying goods.
- Passengers wishing to travel by ship had to discuss and agree on a **8** with its captain.
- The ships travelled frequently on fixed **9** , so arranging a specific journey was relatively easy.
- Passengers on ships usually stayed on the **10**
- Some ships had **11** for those passengers who could afford them.
- The very rich hired a **12** for their privately owned ships.
- **13** could sometimes cause sailings to be cancelled.

READING PASSAGE 2

You should spend about 20 minutes on **Questions 14-26**, which are based on Reading Passage 2 on pages 5 and 6.

Growing more for less

Satellite technology is helping farmers boost crop yields

A For farmers, working out the optimal amount of seed, fertiliser, pesticide and water to scatter on a field can make, or break, the subsequent harvest. Regular laboratory analyses of soil and plant samples from various parts of the field can help – but such expertise is costly, and often unavailable. However, a new and cheaper method of doing this analysis is now on offer. Precise prescriptions for growing crops can be obtained quickly, and less expensively, by calculating the amount of electromagnetic radiation reflected from agricultural land. The data is collected by orbiting satellites.

B Examining the wavelength of the radiation that is reflected can reveal, with surprising precision, the properties of the soil, the quantity of crop being grown, and the levels in those crops of chlorophyll*, various minerals, moisture and other indicators of their quality. If recent and forecast weather data is added to the mix, detailed maps can be produced indicating exactly how, where and when crops should be grown. The service usually costs less than US \$15 per hectare for a handful of readings a year, and can increase yields by as much as 10%.

C Such precision farming using satellite-based intelligence is a relatively new technique. Even so, it is catching on quickly. Five times a year, for example, a French cereal-growers' co-operative called Sevépi purchases satellite data and makes it available to its members in the form of maps of their fields, divided into three or four colour-coded zones per hectare. For each zone, one of about 50 fertiliser formulas is recommended. On top of this, if the wheat in the field has already grown quite high early in the season, and heavy showers are expected, an appropriate dose of growth regulator is recommended for each zone. (Long, fragile stems break more easily in downpours.) Then, farm vehicles equipped with global-positioning system locators automatically mix and apply the prescribed dose to each area.

D France is the pioneer in this sort of surveillance. More farmland is analysed by satellite there than in any other country, according to Infoterra (a subsidiary of EADS Astrium), the firm that is France's largest provider of such information, supplying data to companies such as Sevépi. Moreover, Henri Douche, head of Infoterra's agriculture sales in Toulouse, reckons the amount of monitored farmland will increase as weather patterns change and farmers can no longer rely on the past as a guide to the future. When confounded by the yield variations that these new weather patterns will bring, even farmers who are afraid of new technology will sign up, he says.

E Inexpensive data on the productivity of land is advantageous to governments, too. Areas where fertilisers and pesticides are being applied excessively can be pinpointed, studied and regulated by environmental and land-use agencies. Guy Lafond, an agronomist with Agriculture and Agri-Food Canada, a government agency, says the satellite data it purchases is proving useful for the study of fields with declining productivity in the province of Saskatchewan. Overkill with nitrate fertilisers (which are also a source of greenhouse gases) appears partly responsible. And according to RapidEye, a German satellite operator, insurance companies are also studying satellite data with a view to selling insurance policies to governments of famine-prone countries that might be threatened by crop failure.

F In March, RapidEye began selling data that helps forecast harvests. ‘Too often, farmers limit productivity by managing fields uniformly,’ says Fredrick Jung-Rothenhausler, head of product development at the firm’s headquarters in Brandenburg an der Havel. ‘Our satellites are the first commercial satellites to include the Red-Edge band of the light spectrum, which is sensitive to changes in chlorophyll content. More research will be necessary to realise the full potential of the Red-Edge band. However, this band can assist in monitoring vegetation health, improving species separation and also help in measuring protein and nitrogen content in biomass.’ The company’s data, which comes from both Europe and the Americas, breaks field productivity down into patches just five metres square.

G The advantages that satellite technology provides in terms of precision farming do not have to be restricted to rich countries. In Africa, where many soils have become badly depleted of nutrients, better fertiliser management would greatly improve the situation. As a consequence, the charitable trust World Agroforestry Centre, in the city of Nairobi, in Kenya, has begun to build up a catalogue of the radiation patterns derived from around 100,000 samples of African soils. The information is the key to understanding the potential of these soils to be more agriculturally productive. Once passed on to the International Centre for Tropical Agriculture, based in Colombia, South America, it is intended that the information be used to build a database called the ‘Digital Soil Map’. When ready, this will provide farmers with free forecasts, developed with regularly updated satellite imagery, across farmland in a number of countries in Africa. This is information which will almost certainly assist in improving crop yields. For a hunger-ravaged continent, that is good news indeed.

**chlorophyll: the green substance in plants that allows them to absorb sunlight*

Questions 14 – 20

Reading Passage 2 has seven paragraphs, **A-G**.

Which paragraph contains the following information?

Write the correct letter, **A-G**, in boxes 14-20 on your answer sheet.

NB You may use any letter more than once.

- 14. an example of how farmers in one country are now using satellite data to determine fertiliser use
- 15. a reference to climate change and its effects
- 16. a reference to the effect on the soil of using too much fertiliser
- 17. an example of information that will be shared between different countries
- 18. a mention of the country which is the leader in agricultural technology
- 19. a description of an innovation in satellite imaging which requires further study
- 20. evidence of the cost effectiveness of using satellite technology in agriculture

Questions 21 and 22

Choose **TWO** letters, **A-E**.

Write the correct letters in boxes 21 and 22 on your answer sheet.

Which **TWO** companies obtain information directly from satellites?

- A. Sevépi
- B. Infoterra
- C. Agriculture and AgriFood Canada
- D. RapidEye
- E. World Agroforestry Centre

Questions 23 – 26

Complete the sentences below.

*Choose **NO MORE THAN TWO WORDS** from the passage for each answer.*

Write your answers in boxes 23-26 on your answer sheet.

23. Initially, orbiting satellites are used to measure coming from farmland.

24 believes that confusion about irregular weather will cause more farmers to use satellite technology.

25. As a result of satellite technology, it may become possible to insure against the risk of

26. In Africa, much of the soil suffers from the loss of



READING PASSAGE 3

You should spend about 20 minutes on **Questions 27-40**, which are based on Reading Passage 3 below.

How to win the World Memory Championship

Why do some people seem to have naturally superior memories? Is it talent, a technique or a matter of genetics?

Last August, 34 contestants gathered at the World Memory Championships held at Oxford University. In timed trials, they were challenged to memorise rows of 40-digit numbers, recall the names of 110 people after looking at their photographs and memorise the sequence of a shuffled pack of playing cards. No-one has ever managed this last feat in under 30 seconds and that number has become the target for 'mental athletes'. Even so, most contestants in the memory championships claim to have just average memories and scientific testing confirms that this isn't false modesty. Their feats are based on tricks that capitalise on how the human brain encodes information. We all have the potential to perform just as well if we are taught certain strategies.

Psychologists Elizabeth Valentine and John Wilding recently teamed up with Eleanor Maguire, a neuroscientist at University College London, to study eight of those contestants. They wondered if their brains were different in some way. They put the contestants and a group of control subjects into an MRI* machine and asked them to perform several different memory tests while their brains were being scanned. When it came to memorising sequences of three-digit numbers, the difference between the subjects was, as anticipated, immense. However, when they were shown photographs of magnified snowflakes, images that the contestants had never tried to memorise before, the champions did no better than the control group. When the brain scans were analysed, they showed the champions were activating some brain regions that were different from those the control subjects were using. These regions are known to be involved in visual memory and spatial navigation.

This makes sense when you listen to Ed Cooke, a 23-year-old cognitive science graduate and championship contestant, explain his method. He has memorised a specific person, verb and object that he associated with each card in the deck, and when he commits a shuffled deck to memory, he does it three cards at a time. Every three-card group then forms a single image of a person doing something to an object. He takes the person from the first card, the verb from the second and the object from the third. Yet even meaningful information is hard to remember when there's much of it. That's why memorisers place their images along an imaginary path and why the brain scans of the contestants showed activation in the brain regions associated with visual imagery and navigation in space. Cooke also places images along a path he knows well and when it is time to recall the cards, he takes a mental walk along it and translates the images into cards.

This technique, known as the loci method, reportedly originated in 477 BC with the Greek poet Simonides. He was able to reconstruct a long guest list by closing his eyes and recalling each individual around the dinner table. He had discovered that our brains are exceptionally good at remembering images and spatial information. Evolutionary psychologists have offered a credible explanation: our ancestors found it important to remember where they found their last meal or the way back to shelter. In other words, these skills were really vital.

After Simonides's discovery, the loci method became popular across classical Greece and Rome as a trick for memorising speeches, to make their delivery appear more spontaneous to the audience. Before books, the art of memory was considered a basic of classical education, on a par with grammar and logic. Today external aids supplant natural memory: Palm Pilots track schedules, cell phones store phone numbers, and internet searches turn up a bonanza of information. About the only place where you'll find people still using the loci method is at the championships.

But is there a form of innately superior memory? Valentine and Wilding think so. They divide people with superior memories into two classes: strategists and naturals. The former employ techniques that convert meaningless information into a form the brain can organise and retain, but they tend to be exceptional at memorising only a few types of things. Naturals, on the other hand, show strong memory across a wide variety of tasks and tend to have an easier time holding on to those memories. Apparently, the most famous 'natural' was the 20th-century Russian journalist S. V. Shereshevski, whose memory seemed to have limitless capacity. He could recite strings of numbers learnt decades earlier and recite pages and pages of poetry. He was one of the rare people born with synesthesia, meaning that his senses became intertwined. For example, every number may be associated with a colour or every word with a taste. Synesthetic reactions evoke a response in more areas of the brain, making memory easier.

Nevertheless, other research casts doubt on Valentine's and Wilding's theories. K. Anders Ericsson, a Swedish-born psychologist at Florida State University, thinks anyone can acquire Shereshevski's skills. He cites an experiment with an undergraduate who was paid to take a test of memory called the digit span for one hour a day, three days a week. When he started, he could merely hold seven digits in his head at any given time. But after 250 hours of testing, he stretched to 80. He had developed a strategy for remembering, based on his experience as a professional runner. He associated random numbers with track times. This study led Ericsson to believe that innately superior memory doesn't exist. After reviewing original case studies of naturals, he found that exceptional memorisers were using techniques—sometimes without realising it—and lots of practice.

**MRI (magnetic resonance imaging): a technique for producing images of internal organs*

Questions 27-29

Choose **THREE** letters, **A-G**. The list below includes tasks or activities which require a good memory. Which **THREE** does the writer mention in the article?

- A. learning important historical dates
- B. being able to find a location again
- C. speaking in public
- D. learning grammatical rules by heart
- E. learning words in foreign languages
- F. learning and repeating poetry
- G. learning mathematical formulae

Questions 30-35

Complete the summary below.

Choose **NO MORE THAN THREE WORDS AND/OR A NUMBER** from the passage for each answer.

Competitors at the World Memory Championships aim to memorise cards in a particular order in less than **30 (30)** The **(31)** is an ancient memory strategy still used by competitive memorisers today. Simonides's technique for memorisation relied on the ability to recall visual details and **(32)** People who have a condition known as **(33)** use a combination of senses when they remember something. The student in Ericsson's study was initially able to remember **(34)** only. Ericsson concluded that a combination of **(35)** and strategies were the key to having an excellent memory.

Questions 36 - 40

Do the following statements agree with the views of the writer in Reading Passage 3? In boxes 36 - 40 on your answer sheet, write:

YES *if the statement agrees with the views of the writer*

NO *if the statement contradicts the views of the writer*

NOT GIVEN *if it is impossible to say what the writer thinks about this*

36. Everyone could be capable of completing the same memory tasks as the competitors at the World Memory Championships.

37. The theory that primitive humans developed good memories because of the need to locate food is unconvincing.

38. Logical thinking skills will worsen since people no longer practise memorisation techniques.

39. Valentine and Wilding's beliefs about types of memorisers are open to question.

40. Ericsson's findings agreed with the conclusions of Valentine and Wilding.

